

## TECHNICAL SEMINAR PRESENTATION (PW-EE581)

### TOPIC :Wireless Charging of Electric Vehicles (EVs)

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**ROLL NO. – BTECH/EE/24/059.**

**DATE- 12/O8/2026**

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# INTRODUCTION

## ❑ What is Wireless Charging of Evs ?

- A technology that transfers electrical energy to an electric vehicle battery without a direct conductive cable connection.
- Wireless Power Transfer (WPT) uses electromagnetic coupling between a transmitter pad on the ground and a receiver pad mounted beneath the vehicle.
- Resonant inductive coupling allows efficient transfer across an air gap while maintaining electrical isolation.

## ❑ Why Wireless Charging?

- It improves convenience, supports automated charging, reduces connector handling and can enable future dynamic roadway charging.



# OBJECTIVE

- ❖ Study the operating principle of inductive and resonant wireless power transfer for EVs.
- ❖ Understand the functions of transmitter/receiver coils, compensation capacitors and high-frequency power converters.
- ❖ Analyze the effect of coil misalignment, air gap and coupling coefficient on transferred power and efficiency.
- ❖ Study control, communication, foreign-object detection, thermal management and electromagnetic compatibility.
- ❖ Evaluate advantages, limitations and future applications of stationary and dynamic wireless EV charging.

# CONVENTIONAL VS. WIRELESS CHARGING



# ➤ Block Diagram





# WORKING PRINCIPAL

- Step 1: Sensors continuously read parameters (Voltage, Load Current, Oil Temperature, and Oil Level).
- Step 2: Analog and digital signals are routed to the Node MCU ESP8266 microcontroller.
- Step 3: Node MCU processes raw data, calculates RMS Voltage, RMS Current, Apparent Power, and Temperature values.
- Step 4: Operating parameters are formatted and rendered locally on the 16x2 I2C LCD.
- Step 5: Processing unit checks telemetry against pre-configured safety limits (e.g., Temp > 75 deg)
- Step 6: Node MCU sends telemetry to the IoT Cloud server (Blynk/ThingSpeak) via Wi-Fi.
- Step 7: If a fault condition occurs, the controller updates the cloud alert pin and activates the protective relay.

## About Node MCU ESP8266

### ❑ Microcontroller Overview:

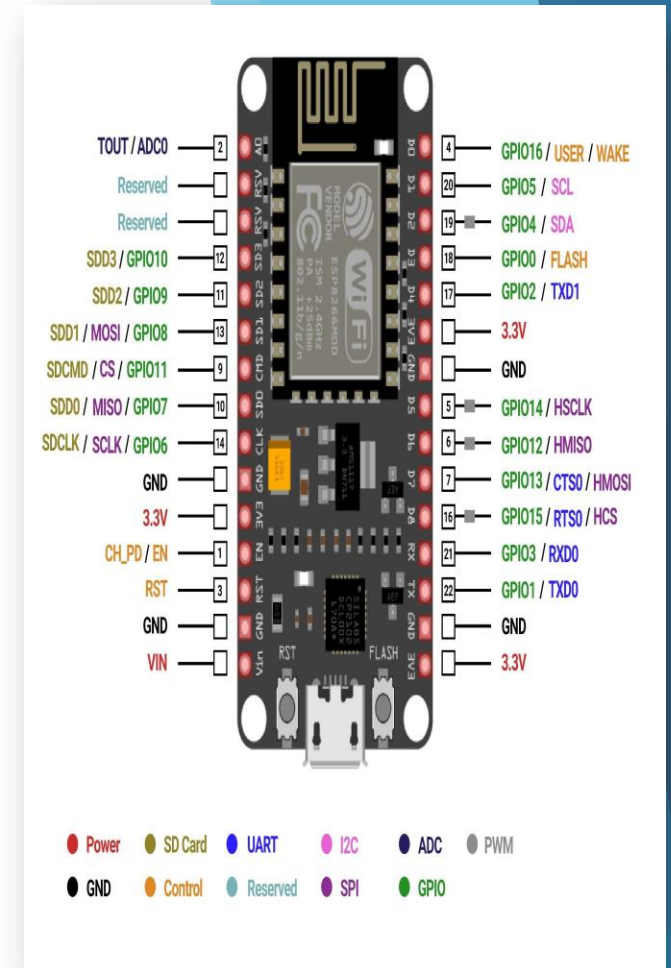
- Low-cost, Wi-Fi-enabled system-on-chip (SoC) based on the ESP8266EX.
- Integrates a 32-bit Tensilica L106 core running at 80 MHz / 160 MHz.

### ❑ Key Specifications:

- Operating Voltage: 3.3 V
- Flash Memory: 4 MB
- Wi-Fi Protocol: IEEE 802.11 b/g/n
- Peripherals: GPIOs, PWM, ADC (10-bit), I2C, SPI, UART

### ❑ Role in System:

- Serves as the central processor for data acquisition, digital signal filtering, local display handling, and TCP/IP wireless transmission.

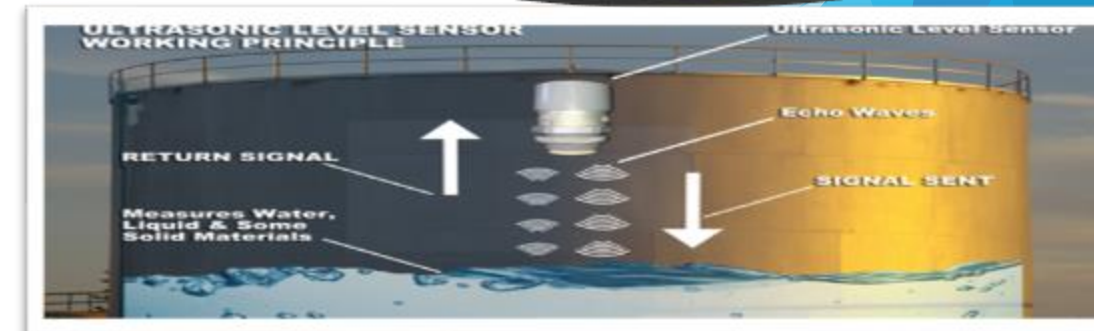
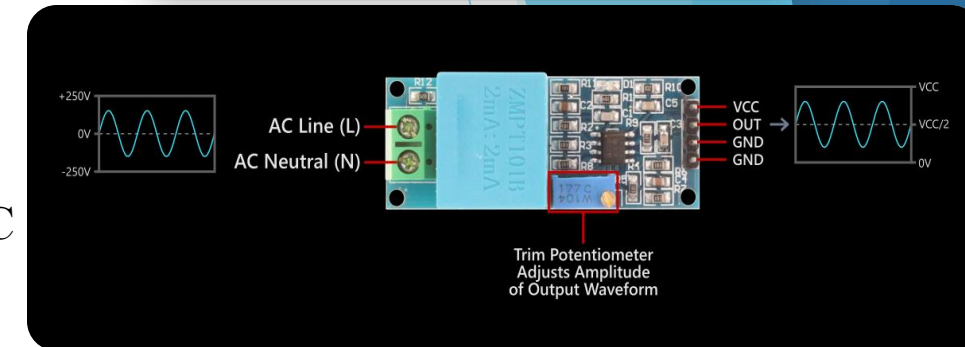
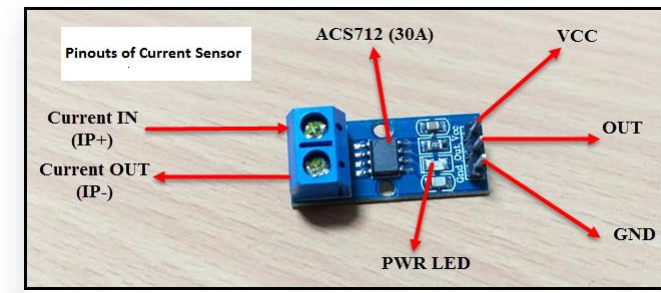


# SENSOR MODULE DETAILS

- ACS712 / CT Sensor: Operates on the Hall Effect principle to sense primary/secondary load current safely.
- ZMPT101B: High-precision voltage transformer module for sampling AC voltage output.

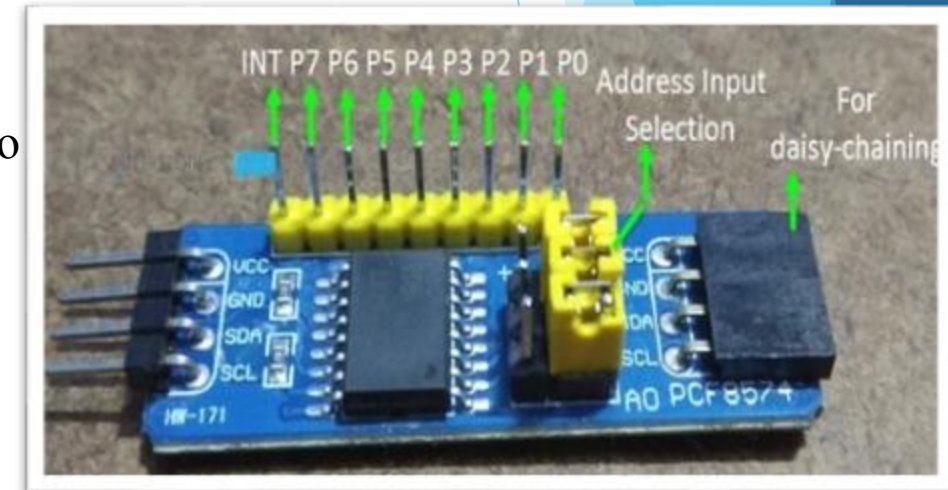
## Temperature & Oil Level Sensing:

- DS18B20 Sensor: Waterproof probe design for immersion in transformer insulating oil; highly accurate 9–12 bit digital output over a One-Wire interface.
- Ultrasonic / Float Sensor: Detects hazardous drop in transformer oil level caused by tank leaks or evaporation.



## About 16x2 LCD with I2C Interface

- **16x2 LCD Character Display**: Displays 2 lines of 16 characters each, providing real-time local monitoring for sub-station operators.
- **PCF8574 I2C Module**:
  - Mounted on the rear of the LCD module to act as an 8-bit I/O expander.
  - Reduces required control lines from 6 or 8 pin connections down to 2 lines (SDA and SCL).
- **Role in System**: Shows real-time readouts of  $V$ ,  $I$ ,  $T$ , and system status (NORMAL / OVERLOAD / FAULT).







## Software & Cloud Platform Used

### ➤ Software Development:

- Arduino IDE: Used for programming C/C++ code, managing board packages, and uploading firmwares.

### ➤ Libraries Integrated:

- ESP8266WiFi.h – Wi-Fi connectivity management.
- LiquidCrystal\_I2C.h – LCD control over I2C.
- OneWire.h & DallasTemperature.h – DS18B20 digital thermal processing.
- BlynkSimpleEsp8266.h or ThingSpeak.h – IoT cloud communication API.

### ➤ Cloud Infrastructure:

- Hosts live dashboards, time-series historical data plotting, and automated push notifications via mobile apps or email alerts.

# Program Structure

# ADVANTAGES& LIMITATIONS

## ➤ Advantages:

- ❖ Continuous Monitoring: Real-time visibility into transformer operating conditions.
- ❖ Predictive Maintenance: Early detection prevents costly winding burnouts and grid shutdowns.
- ❖ Remote Access: Engineers can check health metrics from any location via mobile or web interfaces.
- ❖ Cost Efficiency: Low-cost hardware deployment relative to high transformer replacement costs.

# Real-Life Implementation

- Smart Grid Distribution: Integration into distribution sub-stations for automated asset management.
- Industrial Facilities: Used in large plants to monitor dedicated step-down transformers running critical high-power machinery.
- Remote & Rural Electrification: Reduces physical site inspection trips for isolated transformer stations across wide geographical regions.



# Conclusion

- ❖ The IoT-Based Transformer Health Monitoring System modernizes electrical power distribution through low-cost, continuous, remote condition assessment.
- ❖ By tracking critical metrics like oil temperature, load current, voltage, and oil levels, it provides actionable alerts that mitigate catastrophic asset loss.
- ❖ Implementing this system accelerates the transition toward smart, self-healing power grids.



**THANK YOU!**